Silverline House — 25-Storey Residential HRB

Below Ground Waterproofing Strategy

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1 Introduction

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings. Below-ground waterproofing is provided by a Type C drained protection system in accordance with BS 8102:2022, comprising a primary tanked membrane applied to the external face of the basement retaining walls and underside of the ground floor slab, supplemented by a drainage layer and sump pump system. The membrane is a hot-applied reinforced bituminous sheet system with a minimum thickness of 4 mm applied to prepared concrete substrate. Lap seals are formed using a heat torch and are inspected and pressure-tested prior to backfilling. The drainage layer consists of a geocomposite drainage sheet with a geotextile filter fabric bonded to the soil-facing surface, discharging to perimeter drainage sumps. Penetrations through the tanking system for services entries are sealed using proprietary puddle flanges and sleeve systems with BBA certification.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 System Specification and Installation

Below-ground waterproofing is provided by a Type C drained protection system in accordance with BS 8102:2022, comprising a primary tanked membrane applied to the external face of the basement retaining walls and underside of the ground floor slab, supplemented by a drainage layer and sump pump system. The membrane is a hot-applied reinforced bituminous sheet system with a minimum thickness of 4 mm applied to prepared concrete substrate. Lap seals are formed using a heat torch and are inspected and pressure-tested prior to backfilling. The drainage layer consists of a geocomposite drainage sheet with a geotextile filter fabric bonded to the soil-facing surface, discharging to perimeter drainage sumps. Penetrations through the tanking system for services entries are sealed using proprietary puddle flanges and sleeve systems with BBA certification. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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3 Inspection and Testing

A third-party inspector appointed by the waterproofing contractor will inspect all membrane installation, lap seals, and penetration details prior to backfilling. Inspection records will be retained as part of the Golden Thread documentation. A site investigation was carried out in accordance with BS EN 1997-2 and BS 5930. The investigation comprised twelve rotary core boreholes to a maximum depth of 30 m below existing ground level, together with in-situ standard penetration tests at 1.5 m intervals. Laboratory testing was carried out on selected samples to determine soil classification, shear strength parameters, consolidation characteristics, and chemical contamination. The ground conditions comprise approximately 3 m of made ground overlying Terrace Gravels (3–9 m depth), London Clay (9–22 m), and Lambeth Group deposits below 22 m. Groundwater was encountered at 2.5 m depth in the gravels. The design groundwater level for structural purposes is taken as 0.5 m below existing ground level.

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